Relation between degrees and radians

We know that the circumference of a circle of radius 1unit is 2π. One complete revolution of the radius of unit circle subtends 2π radians. A circle subtends at the centre an angle whose degree measure is 360°.

2π radians $= 360^\circ$ $\quad \pi$ radians $= 180^\circ$

$$1\text{radian} = \frac{180^\circ}{\pi}$$

Example 4.1

Convert: (i) 160° into radians

(ii) $\frac{4\pi}{5}$ radians into degree

(iii) $\frac{1}{4}$ radians into degree

Solution

i) $160^\circ = 160 \times \frac{\pi}{180} radians = \frac{8}{9}\pi$

ii) $\frac{4\pi}{5} radians = \frac{4}{5}\pi \times \frac{180^\circ}{\pi} = 144^\circ$

iii) $\frac{1}{4} radian = \frac{1}{4}\frac{180}{\pi}$

$= \frac{1}{4} \times 180 \times \frac{7}{22} = 14^\circ 19' 5''$

4.1 Trigonometric Ratios

4.1.1 Quadrants:

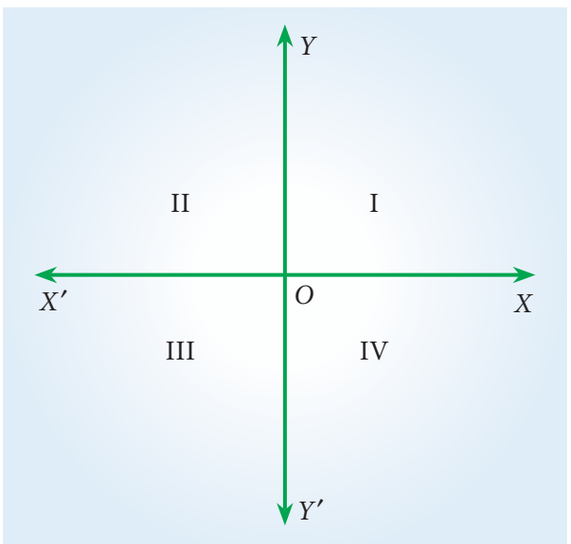


Fig. 4.3

Let $X'OX$ and $Y'OY$ be two lines at right angles to each other as in the figure. We call $X'OX$ and $Y'OY$ as X axis and Y axis respectively. Clearly these axes divided the entire plane into four equal parts called "Quadrants".

The parts XOY, YOX', $X'OY'$ and $Y'OX$ are known as the first, second, third and the fourth quadrant respectively.

Example 4.2

Find the quadrants in which the terminal sides of the following angles lie.

(i) -70° (ii) -320° (iii) 1325°

Solution (i)

(i) The terminal side of -70° lies in IV quadrant.

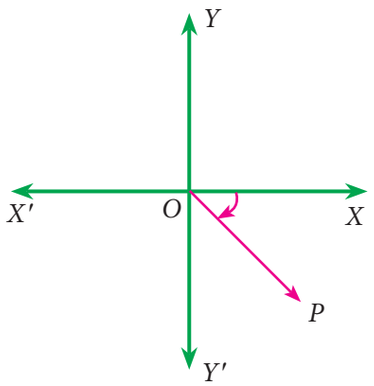


Fig. 4.4

(ii) The terminal side -320° of lies in I quadrant.

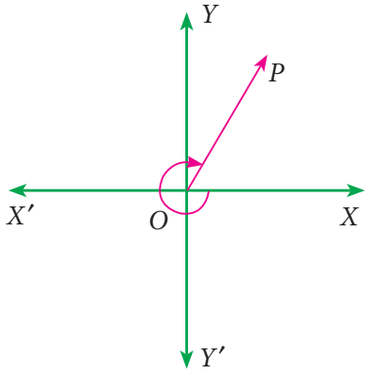


Fig. 4.5

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